Nuo Xu

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EDUCATION

Johns Hopkins University (Baltimore, US) **09/2017-Present**
Whiting School of Engineering, Department of Electrical and Computer Engineering

- Master candidate of Electrical and Computer Engineering
- **GPA:** 3.7/4.0
- **Main Courses:** Digital Image Processing; Algorithms for Sensor-Based Robotics; Robotic Devices, Dynamic and Kinematic; Machine Learning for Signal Processing, Intro to Algorithms, Computer Vision, Audio and Vision Signal Processing

Tianjin University (Tianjin, China) **09/2013-07/2017**
Department of Electrical and its Automation Engineering

- Received Bachelor's Degree of Automation
- **GPA:** 3.71/4.0; **Rank:** 15/168
- **Main Courses:** Auto-control Theory, C++, Signal Analysis and Processing, Electric Power and Electronics, Motion Control Theory, Computer Control Technology, Process Control System, Pattern Recognition, Single Chip Processor, FPGA, Industrial Control Network
- **Honor:** College of Electrical Engineering & Automation, 2015 School-level Merit Student (10%) & Scholarship (one spot); 2016 School-level outstanding cadres of College of Electrical Engineering & Automation (10%); Outstanding graduates Awards (5 spots for 17 candidates)

ACADEMIC EXPERIENCES

Satellite Surgery **06/2018-08/2018**
Director: *Prof. Simon Leonard;*

- Programmed the Cartesian path planner with collision detection in *MoveIt!* for UR10 in order to cut the insulation layer of satellite alone a given path. UR10 with cutter at end-effector will be used to dock with the satellite, cut and peel back the insulation layer
- Designed a visualization model in *MoveIt!* where the cutter of UR10 can only detect collision with satellite. The obstacle configuration, mesh of satellite and waypoints on Cartesian path can load from json file
- With RVinci plugin, RViz Interaction and Navigation with da Vinci, the path generated from planner can be viewed in stereo image. Points can also be picked in RViz with the gripper of da Vinci from the stereo image

Hopkins Delivery **09/2017-05/2018**
Director: *Prof. Charbel Rizk;* ***Position:*** *Software and System Engineer;*

- The goal of the project is to demonstrate the concept of drone delivery system throughout the JHU Homewood campus. The delivery system is built upon DJI Matrice 100, ZED stereo camera, NVIDIA Jetson TX2 and Arduino
- Programmed the collision avoidance algorithm with depth image captured by ZED stereo camera
- Programmed a ROS package for communication between ultrasound sonars MaxSonar MB1204, electromagnetics and Arduino & NVIDIA Jetson TX2 with rosserial
- Programmed the landing algorithm. The drone will land on specified tag on the ground by detecting position the Apriltags with downward camera
- Demo Video: https://vimeo.com/256362144

Visualization System of Acupuncture Manipulation based on Augmented Reality **02/2017-06/2017**

- Designed a platform with Verilog HDL on the FPGA to pick up images and transferred the image data to host computer
- Programmed a visualized algorithm to realize human acupuncture points registration on a real human body picture
- A patent, *The visualization platform of acupuncture and moxibustion based on Augmented Reality and human acupoint registration algorithm*, was applied on April 11th, 2017. This patent built a platform for the visualization of acupuncture and moxibustion based on Augmented Reality. The visualization platform can accurately register the location of human acupoint to the real human body virtually. The application number is cn 201710231105.1.

PRACTICAL EXPERIENCES

Course Assistant **09/2018-12/2018**

Department: *Computer Science;* ***Position:*** *part-time Course Assistant; 10 hours/week*

- Assisted with grading and answered questions
- Programmed Ground Robot Jackal for Students' Assignments

Tianjin University Integrative Service Administration Office **03/2017-06/2017**

Department: *International Department;* ***Position:*** *part-time Office Assistant; 10 hours/week*

- Data entry, scanning, photocopying, file management, light internet research assistance

Tianjin Career Training Center **08/2016-09/2016**

- Designed a water tank control system with SIEMENS PLC which can maintain the water level at the setting level
- Built a system which can collect, transfer and classify adjunctions for grinding wheels with PLC S7-200
- Established an easy Pneumatic food package automation production line

Tsinghua University **07/2016-08/2016**

Department: *Microelectronics and Nano-electronics;* ***Position:*** *Intern; 40 hours/week*

- Acquired essential program skill, especially about the widely-used OpenCV for computer vision
- Searched essays about project and made criterions for the project paper by referring these essays
- Debugged algorithm and self-learned STM microchip

COURSE PROJECT

Alexabot **08/2018-11/2018**

- Alexabot is the collaboration system between Amazon Alexa and Turtlebot; System includes hand gesture control, Face ID, voice control via Alexa Skill and web-teleop control
- Built the hand gesture control with Convolutional Neural Network and Naïve Bayes; Built the json2ros nodes, teleop-voice control node and turtlebot following node with Microsoft Kinect
- Demo video https://www.youtube.com/watch?v=TXflYI3eTqI&t=7s

Augmented Reality Drum Game **08/2018-11/2018**

- Built an AR Drum Game with OpenCV and OpenG. Game can response to two hitting methods: vertically hitting and horizontally hitting. Each hitting methods correspond to a colored circle; When players correctly hit the circle with required hitting method, points will be earned
- Demo video https://www.youtube.com/watch?v=bi6BYpFSOGs&t=24s

Pick-and-Mark and Draw portrait with UR5 **05/2018**

- Used the UR5 robot to perform a task "place-and-mark". Implemented a program that allows you to first taught the UR5 the position of the start and target locations and then automatically completed the place-and-mark operation and several different types of control(Inverse kinematics, Rate control using differential kinematics, Gradient-based control)
- Processed the image with color gradient method, the manipulation control method is Inverse Kinematics Control. Let the UR5 firstly go to the location which is vertically 10cm above the taught location for pen, then the UR5 gradually goes down to mark a point on paper. The whole face consists of more than 2,000 points. UR5 draws point by point. After all the points are drawn, they form Professor's face.
- Demo video https://www.youtube.com/watch?v=OsZfEkdaMbs&t=4s

Sampling-based Path Planner of a UR5 robot **03/2018**

- Used the *MoveIt!* Libraries and implement a sampling-based path planner to plan collision-free motions of a UR5 robot
- Entered the robot arm through the door of a car frame and successfully positioned the end-effector near the steering wheel within 30 seconds.

ADDITIONAL INFORMATION

- **Software:** ROS, Proteus, Git, Visual Studio, Linux, OpenCV, Mathmatica
- **Hardware:** PLC, Arduino, FPGA, Particle Photon
- **Programming Skills:** C++, C, Python, Verilog HDL, MATLAB, Java, html5, JavaScript